

A/B Testing: Mobile Game Retention Experiment

Does moving the first progression gate from Level 30 to Level 40 affect player retention?

1. Project Overview

Project Objective

This project evaluates whether moving the first progression gate in a mobile game from **Level 30 to Level 40** affects player retention. An A/B testing framework compares two randomly assigned groups:

- **Control:** Gate at Level 30
- **Treatment:** Gate at Level 40

The primary outcomes are **Day-1 retention** and **Day-7 retention**. The analysis goes beyond simple percentage comparisons by using statistical hypothesis testing, confidence intervals, bootstrap resampling, and practical business impact calculations.

2. Business Problem

The product team is considering moving the first progression gate from Level 30 to Level 40. The underlying product question is:

Does allowing players to progress further before encountering the first gate improve player retention?

A higher gate could allow players to experience more gameplay before encountering a restriction. However, it could also change player progression and engagement in unexpected ways. The experiment was therefore designed to determine whether the change produced a measurable difference in player retention.

3. Experiment Design

Players were randomly assigned to one of two groups.

Group	Experience	Role
gate_30	First gate at Level 30	Control
gate_40	First gate at Level 40	Treatment

The experiment contained approximately **90,000 players**. Initial group distribution:

Group	Players	Share
Gate 30	44,700	49.56%
Gate 40	45,489	50.44%
Total	90,189	100%

The approximately 50/50 split provides a reasonable basis for comparing the two experiences.

4. Dataset

The dataset contains four primary variables used in the analysis.

Column	Description
userid	Unique player identifier
version	Experiment group: gate_30 or gate_40
sum_gamerounds	Total game rounds played
retention_1	Whether the player returned on Day 1
retention_7	Whether the player returned on Day 7

The retention variables are Boolean (True = player returned, False = did not return). Because Boolean values behave as 1/0 in Python, their mean gives the retention rate. For example, `df["retention_1"].mean()` returns the proportion of players retained on Day 1.

5. Data Understanding

Initial checks included dataset dimensions, column names, data types, missing values, duplicate users, unique users, game-round statistics, retention distributions, and experiment group counts.

Initial dataset

- **90,189 rows, 90,189 unique users, 0 duplicate user IDs**
- No negative game-round values
- Initial overall retention: Day-1 **44.52%**, Day-7 **18.61%**

6. Randomisation and Group Balance

Before comparing retention, it is important to verify that the two groups are reasonably comparable. Initial gameplay statistics:

Metric	Gate 30	Gate 40
Users	44,700	45,489
Mean rounds	52.46	51.30
Median rounds	17	16
Std. Dev.	256.72	103.29

The control group's standard deviation and mean were unusually high because of an extreme observation, so the data was investigated before continuing.

7. Data Quality: Anomaly Detection

One player recorded an extremely unusual number of game rounds:

Field	Value
User ID	6390605
Group	Gate 30

Field	Value
Game rounds	49,854

The next-highest game-round value was only **2,961**, so the observation was treated as an obvious data anomaly.

Action taken

```
df_clean = df[df["sum_gamerounds"] != df["sum_gamerounds"].max()].copy()
```

Before cleaning: **90,189 rows**. After cleaning: **90,188 rows**. Rows removed: **1**. The purpose was not to remove a player because they were inconvenient to the analysis, but because the observation was clearly inconsistent with the rest of the gameplay distribution and heavily distorted summary statistics.

8. Group Balance After Cleaning

Metric	Gate 30	Gate 40
Users	44,699	45,489
Share	49.56%	50.44%
Mean rounds	51.34	51.30
Median rounds	17	16
Std. Dev.	102.06	103.29
Maximum rounds	2,961	2,640

Conclusion

The control and treatment groups are approximately balanced in both sample size and gameplay behavior. This supports proceeding with the retention comparison.

9. Day-1 Retention Analysis

Group	D1 Retention
Gate 30	44.82%
Gate 40	44.23%

Difference (Gate 40 - Gate 30): **-0.59 percentage points**. The Gate 40 group had an observed Day-1 retention rate about 0.59 pp lower. At this stage this is only an observed difference; statistical testing is required to determine whether it could reasonably be due to random variation.

10. Day-7 Retention Analysis

Group	D7 Retention
Gate 30	19.02%
Gate 40	18.20%

Difference (Gate 40 - Gate 30): **-0.82 percentage points**. The Gate 40 group therefore had approximately 0.82 pp lower Day-7 retention.

11. Hypothesis Testing

Separate tests were performed for Day-1 and Day-7 retention.

- **Null hypothesis:** There is no difference in retention between Gate 30 and Gate 40.
- **Alternative hypothesis:** There is a difference in retention between Gate 30 and Gate 40.

A **two-proportion z-test** was used because retention is a binary outcome and the experiment contains two independent groups. The significance level was **alpha = 0.05**, corresponding to a 95% confidence level.

12. Day-1 Statistical Test

- z-statistic: **1.787**
- p-value: **0.0739**
- 95% CI: **-1.24 pp to +0.06 pp**

Because $0.0739 > 0.05$, there is insufficient evidence to conclude that Gate 40 changed Day-1 retention at the 5% significance level. The confidence interval also includes zero.

Interpretation

The data is compatible with a small negative effect, essentially no effect, or a very small positive effect. The observed -0.59 pp difference should not be treated as a confirmed treatment effect.

13. Day-7 Statistical Test

- z-statistic: **3.157**
- p-value: **0.00159**
- 95% CI: **-1.33 pp to -0.31 pp**

Because $0.00159 < 0.05$, there is statistically significant evidence of a difference between the two groups. The entire confidence interval is below zero.

Interpretation

The data provides strong evidence that Day-7 retention was lower for the Gate 40 group. The plausible treatment-control difference is approximately -1.33 to -0.31 percentage points, with the observed difference being -0.82 percentage points.

14. Confidence Intervals

Metric	Observed Difference	95% CI
D1	-0.59 pp	-1.24 to +0.06 pp
D7	-0.82 pp	-1.33 to -0.31 pp

- **D1:** The interval crosses zero, so no statistically supported difference is established.
- **D7:** The interval remains below zero, so the data supports a negative D7 retention difference for Gate 40.

15. Bootstrap Analysis

To further evaluate the uncertainty around the estimated retention differences, a bootstrap procedure was performed:

1. Take the control group's retention observations.
2. Randomly resample with replacement.
3. Take the treatment group's retention observations.
4. Randomly resample with replacement.
5. Calculate Treatment - Control retention.
6. Repeat **10,000 times**.
7. Examine the resulting distribution.

This provides an empirical view of the range of retention differences that could arise from repeated samples.

16. Day-1 Bootstrap

- Mean bootstrap difference: **-0.594 pp**
- 95% bootstrap CI: **-1.253 pp to +0.048 pp**

The bootstrap interval crosses zero. **Conclusion:** the bootstrap agrees with the two-proportion z-test; there is insufficient evidence of a statistically significant Day-1 retention difference.

17. Day-7 Bootstrap

- Mean bootstrap difference: **-0.819 pp**
- 95% bootstrap CI: **-1.326 pp to -0.317 pp**

The entire bootstrap interval remains below zero. **Conclusion:** the bootstrap independently supports the finding that Gate 40 has lower Day-7 retention, and is very close to the analytical confidence interval:

Method	95% CI
Analytical CI	-1.326 to -0.310 pp
Bootstrap CI	-1.326 to -0.317 pp

This consistency increases confidence in the estimated D7 effect.

18. Practical Business Impact

Statistical significance tells us whether the difference has strong evidence behind it. The product team also needs to understand the **scale** of the difference. The D7 difference was **-0.818 percentage points**. For 1 million players:

$$-0.008183 \times 1,000,000 = \text{approx. } -8,183 \text{ players}$$

If the same observed difference occurred among 1 million players, approximately **8,183 fewer players would be retained on Day 7 under Gate 40**. For D1:

$$-0.005915 \times 1,000,000 = \text{approx. } -5,915 \text{ players}$$

However, because the D1 result was not statistically significant, this should be presented as an **observed difference**, not a confirmed treatment-caused loss.

19. Statistical vs Practical Significance

Statistical significance: "Do we have enough evidence that a difference exists?" For D7: **Yes**.

Practical significance: "Is the difference large enough to matter to the business?" For D7, the observed difference is 0.82 pp, about 8,183 players per million. Whether that is commercially important depends on:

- Revenue per retained player
- Lifetime value
- Advertising revenue
- Acquisition cost
- Long-term retention
- Engagement

Statistical significance alone should not determine the final product decision.

20. Limitations

20.1 Revenue was not measured

The experiment focuses on retention and does not directly show whether Gate 40 affects revenue, ARPU, LTV, purchases, or advertising revenue.

20.2 Longer-term retention

Only D1 and D7 were analyzed. Additional measurements should include D14, D30, and potentially D60.

20.3 Novelty effect

Players may initially behave differently because the new progression system is unfamiliar. The observed effect could change over a longer period.

20.4 Single experiment

One experiment provides evidence for this tested population and period. Repeating the experiment or testing another cohort can provide additional evidence about whether the result persists.

20.5 Player segments

The overall average may hide differences between player groups. Further analysis could examine highly engaged vs casual players, new vs returning users, acquisition channels, geographic segments, and gameplay intensity.

21. Product Decision

The experiment does not provide evidence that moving the gate from Level 30 to Level 40 improves retention. The strongest result is the Day-7 retention difference of **-0.82 percentage points** ($p = 0.00159$; 95% CI = -1.33 to -0.31 pp), also supported by the 10,000-iteration bootstrap.

Decision

The Level 40 configuration should not be adopted as the default based on retention results alone. Further analysis should be conducted before making a final product decision.

22. Recommended Next Steps

Step	Focus
1. Longer-term retention	Track D14, D30, D60.
2. Monetisation	Measure revenue per user, ARPU, LTV, purchase conversion, advertising revenue.

Step	Focus
3. Engagement	Measure game rounds, sessions, session duration, progression speed.
4. Player experience	Investigate player feedback, drop-off around the gate, progression friction, gameplay satisfaction.
5. Segmentation	Determine whether the effect differs by player engagement, acquisition source, geography, or player type.
6. Follow-up experiment	Repeat the experiment with another cohort to determine whether the D7 effect persists.

23. Final Executive Summary

The A/B test evaluated whether moving the first progression gate from **Level 30 to Level 40** affected player retention across approximately **90,000 players**.

The experiment groups were approximately 50/50 balanced. One extreme observation containing **49,854 game rounds** was removed as an obvious anomaly, after which the groups remained well balanced.

Day-1 retention was **44.82% for Gate 30** versus **44.23% for Gate 40**, a difference of **-0.59 pp**. This result was not statistically significant (**p = 0.0739**).

Day-7 retention was **19.02% for Gate 30** versus **18.20% for Gate 40**, a difference of **-0.82 pp**. This result was statistically significant (**p = 0.00159**), with a 95% confidence interval of approximately **-1.33 to -0.31 pp**.

A **10,000-iteration bootstrap** produced a similar D7 interval of approximately **-1.33 to -0.32 pp**, providing additional support. At a scale of **1 million players**, the observed D7 difference corresponds to approximately **8,183 fewer Day-7 retained players** under Gate 40.

The experiment therefore does not show a retention improvement from moving the gate to Level 40. Before making a final product decision, the team should evaluate longer-term retention, monetisation, engagement, player experience, and repeat the experiment across another cohort.

24. Project Skills Demonstrated

Area	Skills
Data Analytics	Data cleaning, exploratory data analysis, group comparison, distribution analysis, outlier/anomaly detection
Statistics	A/B testing, hypothesis testing, two-proportion z-test, p-values, confidence intervals, statistical and practical significance, bootstrap resampling
Python	Pandas, NumPy, SciPy, Statsmodels, Matplotlib
Product Analytics	Experiment design, retention analysis, business impact estimation, product decision-making, risk assessment, follow-up metric design
Communication	Executive summary, PM decision memo, statistical findings translated into business language

25. Project Conclusion

The experiment found no statistically significant Day-1 retention difference after moving the progression gate from Level 30 to Level 40. However, Day-7 retention was 0.82 percentage points lower for the Level 40 group, with strong statistical evidence supporting the difference. At 1 million players, this represents

approximately 8,183 fewer Day-7 retained users based on the observed effect. The result was also confirmed through 10,000 bootstrap iterations. Further testing of longer-term retention, revenue, engagement, and player experience is required before making a final product decision.